NBA Player Scoring Analysis

STAT 3675

Data Description

Data source: Basketball Reference (2025–2026 season)

Observations: 100 NBA players

Unit: Individual player

Response variable: Points per game (PTS)

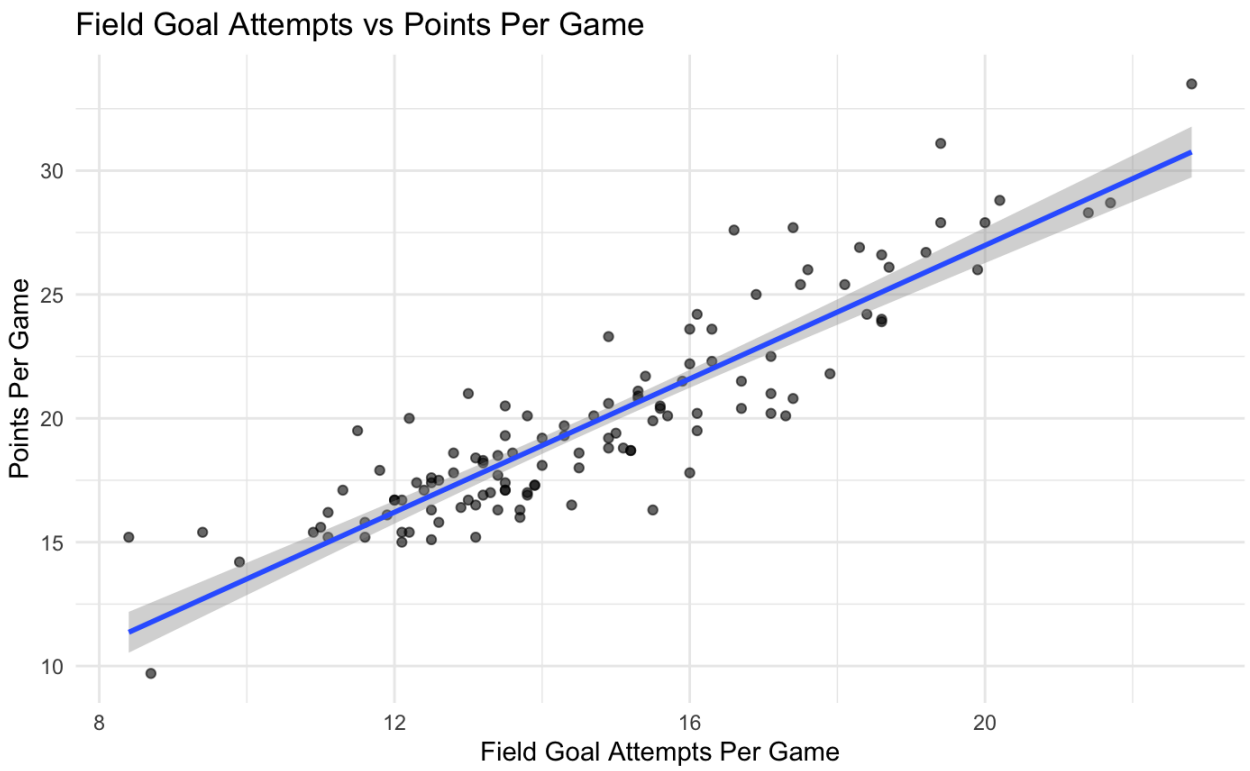
Predictors:

- Minutes (MP)
- Field goal attempts (FGA)
- 3PT attempts (3PA)
- Free throw attempts (FTA)
- Assists (AST), rebounds (TRB), turnovers (TOV)

Research Question:

- Which player statistics best predict points per game?

Data Cleaning & Exploratory Analysis



Process:

- Removed duplicates and cleaned dataset
- Converted variables to numeric
- Handled missing values

Insights:

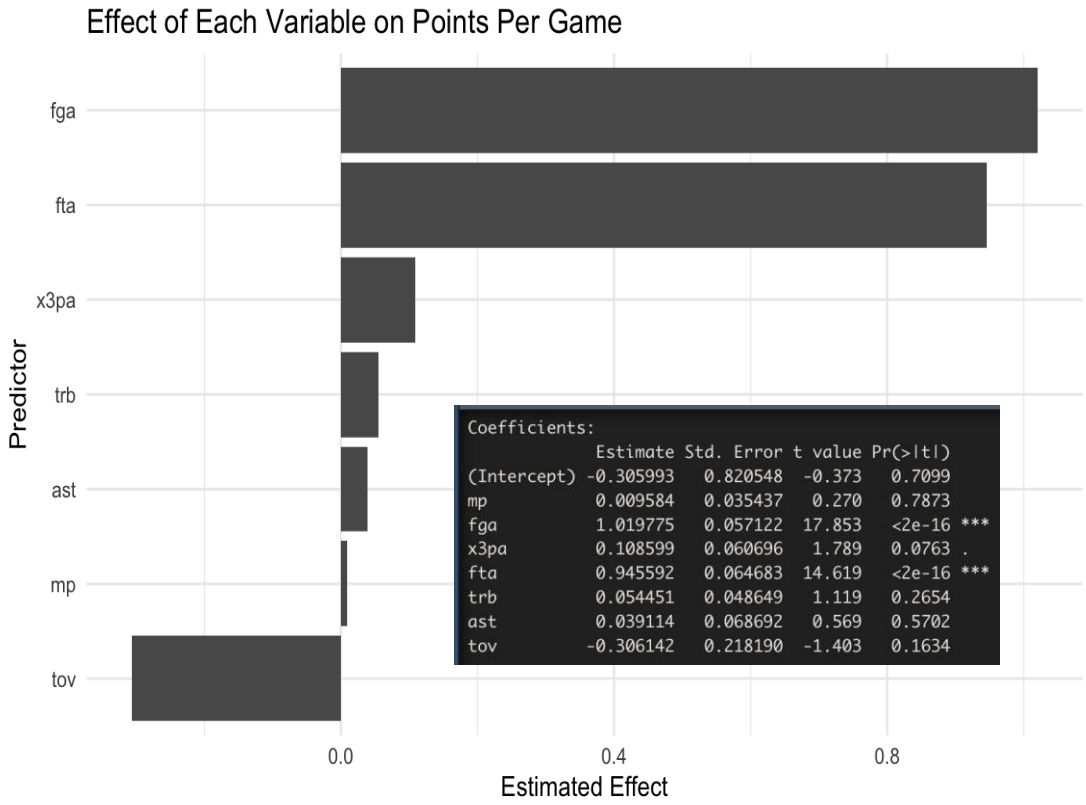
- Strong positive direct relationship: **FGA and PTS**
- Minutes related to scoring
- Linear pattern supports regression

Multiple Linear Regression Model Results

Equation:

PTS ~ MP + FGA + 3PA + FTA + TRB + AST + TOV

- FGA = strongest predictor of scoring
- MP (minutes) significantly increases PTS
- FTA also contributes to scoring
- Assists/rebounds less direct effect

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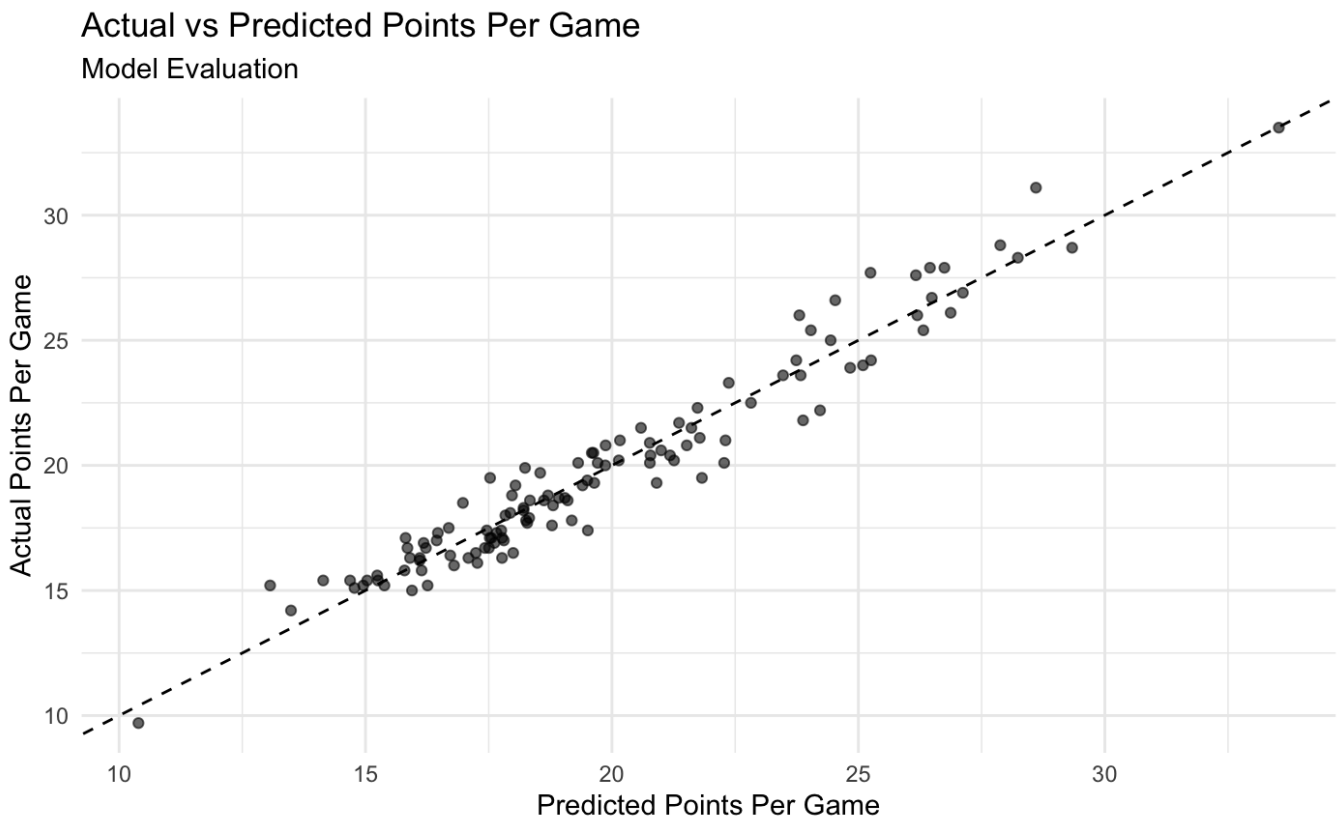
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.305993   0.820548  -0.373   0.7099
mp           0.009584   0.035437   0.270   0.7873
fga          1.019775   0.057122  17.853   <2e-16 ***
x3pa         0.108599   0.060696   1.789   0.0763 .
fta          0.945592   0.064683  14.619   <2e-16 ***
trb          0.054451   0.048649   1.119   0.2654
ast          0.039114   0.068692   0.569   0.5702
tov         -0.306142   0.218190  -1.403   0.1634
```

MLR Model Evaluation

ACTUAL VS PREDICTED PLOT

- Model predictions align with actual values
- Residuals centered around 0
- Some variability at high scoring levels
- Multicollinearity present resulting in justified ridge



Conclusion and Limitations

Scoring is primarily driven by shot attempts (FGA), free throws (FTA), and overall opportunity.

Limitations:

- Observational data → no causation
- Per-game averages (no game-level detail)
- Missing context (defense, role, injuries)
- No time structure (home/away, playoffs)

Future Work:

- Use advanced stats (efficiency, usage rate)
- Include game-level or team-level data